Conceptual Replication of Feasibility Study and Machine Learning Modeling for Menstrual Cycle Prediction Using mcPHASES Data

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Abstract

Background: The menstrual cycle reflects fundamental aspects of women's reproductive and general health, yet traditional tracking tools rely solely on self-reported data. Very few studies have validated the use of consumer wearables across diverse populations for accurate Menstrual phase prediction.

Methods: Our two-phase study utilizes the Canadian mcPHASES dataset, which contains multimodal data from Fitbit sensors, glucose monitors, daily hormone measurements, and self-reported symptoms. Phase 1 focuses on a conceptual replication of a similar German study (FEMFIT) using mixed-effects models to assess the feasibility of the study and identify patterns across menstrual phases. Phase 2 involved the development of machine learning models for accurate menstrual phase prediction.

Results: The Canadian cohort proved to have superior retention (95% through 12 weeks) and data completeness (~97% valid wearable data) compared to the German reference study. Mixed-effects models replicated previous physiological findings: resting heart rate showed significant variation across the four Menstrual cycle phases. However, PMS-stratified activity patterns did not replicate, suggesting behavioral measures may be influenced by regional differences. For phase prediction, XGBoost with temporal features achieved high performance (~71% accuracy, 0.9 AUC), with the 3-day rolling average of nightly body temperature proving to be a critical predictor of phase.

Conclusions: Physiological data obtained from wearable devices, especially resting heart rate and body temperature trends, demonstrate robust cross-regional validity for menstrual cycle predictions. Meanwhile, behavioral and subjective measures show greater variation between populations. These findings validate

the feasibility of consumer wearables for menstrual health monitoring across different populations. Efficient, privacy-preserving algorithms can achieve significant results without the need for comprehensive data collection.

1. Introduction

The menstrual cycle is a fundamental biological process shaped by a tightly regulated interplay of hormones: Estrogen, Progesterone, Luteinizing Hormone (LH), and Follicle-Stimulating Hormone (FSH) that modulate physiological systems across four distinct phases: Menstrual, Follicular, Ovulatory, and Luteal. Each phase exhibits a characteristic hormonal signature and predictable physiological changes, influencing basal body temperature, heart rate, sleep, and emotional states. Because these physiological rhythms are phase-dependent, accurate cycle-phase characterization is both clinically and personally valuable.

Women track their cycles for diverse purposes, including fertility planning, anticipating menstrual discomfort, and broader lifestyle optimization. The widespread use of digital diaries and period-tracking apps has accelerated this shift toward personalized reproductive health awareness. Yet contemporary tracking tools rely on self-reported heuristics, resulting in imprecise predictions and motivating demand for scientifically grounded methods that leverage physiological data from consumer wearables.

Additionally, despite the recent growth of FemTech and the availability of large-scale tracking data, the existing scientific literature suffers from several major limitations. First, many studies depend exclusively on self-reported symptoms or cycle dates, often omitting direct physiological measurements. Second, datasets with continuous physiological signals, hormone measurements, and symptom reports remain extremely scarce, especially in studies related to female physiology. Third, many large-cohort studies involve predominantly U.S. or European app users, with limited representation of specific regional populations, including Canadian women. Critically, the few studies that combine female hormones with wearable data rarely support external or cross-regional validation. These constraints create a central

problem: there is still a lack of validated, generalizable, and physiologically grounded models for menstrual phase prediction, particularly in data-scarce or region-specific contexts.

High-quality, publicly available datasets such as menstrual cycle Physiological, Hormonal, and Self-Reported Events and Symptoms (mcPHASES)[1] help bridge this gap. It provides a combination of modalities: continuous wearable-like signals, and daily hormonal ground truth enabling physiological analysis and predictive modeling in data-scarce settings.

This study also incorporates the Feasibility Study on Menstrual Cycles With Fitbit Device (FEMFIT)[2] dataset, a German cohort with comparable wearable-derived measures but without hormonal data. It enables conceptual replication of previously published mixed-effects analyses but does not support machine learning (ML) model development. Accordingly, all ML models in this study are developed and evaluated solely on the Canadian mcPHASES dataset, which contains the full multimodal information required for cycle-phase prediction.

The combination of these datasets fills a critical gap in the literature by assessing cross-regional stability in phase-dependent physiology and evaluating multimodal signals for accurate ML-based menstrual phase prediction in the Canadian population.

To address this need, this study has two objectives:

1. Conceptual Replication for Feasibility Analysis and Cross-Regional Signal Validation: We replicated the analytical framework of a German *JMIR mHealth* mixed-effects framework using the Canadian mcPHASES dataset to assess cross-cohort consistency of phase-specific physiological patterns and the feasibility of wearable-based menstrual tracking in a distinct population.

2. Advanced Machine Learning Model Development (Canadian Cohort Only): Leveraging the multimodal nature of mcPHASES, we developed ML models to capture stable predictive relationships across wearable-derived signals, hormone trajectories, and glucose values. This approach enables day-specific

prediction of menstrual phases, fertility window timing, and menstruation onset in the only dataset where hormonal information is available.

This work aims to contribute to the limited availability of research on female hormone dynamics and menstrual cycle evaluation, particularly studies employing machine learning approaches.

2. Methods

2.1. Study Design

This study is structured as a two-phase investigation; a conceptual replication of an established analytical framework followed by an extended analysis using Machine Learning (ML) models. Since the PhysioNet mcPHASES dataset is relatively small, the focus remains on methodological reproducibility and proxy feasibility rather than large-scale regional generalization.

2.1.1. Phase 1: Conceptual Replication for Feasibility Analysis

In a conceptual replication, the researcher tests the same theoretical idea as the reference study, but uses different methods, measures, or operationalizations.[3] Simons emphasized in 2017 that conceptual replications probe the limits and generality of a theoretical effect by using different samples, contexts, and methods.[4] A study qualifies as a conceptual replication when it[3] tests the same underlying theoretical relationship[4], applies different samples, measures, or data sources, evaluates whether the same theoretical pattern emerges, and contributes to understanding the generalizability of the original theory. [3,4,5,6] In addition, there are three characteristics of a replication study, according to Etienne: methodology similarity, replication differences, and investigator independence.[7]

In this study, the primary purpose is to assess both the feasibility and consistency of using the Fitbit physiological signals to capture variations in the menstrual cycle. This aligns with the core theoretical relationship of the original claim that physiological signals experience systematic changes across cycle

phases. This replication was conducted by an independent research team with no prior involvement in the original study, enhancing the objectivity and generalizability of the findings. The data sources are different (mcPHASES is from Canada, the data in the reference study is from Germany), which fits the relocation differences. The sample size, average age of samples, and data collection time are consistent with the reference study, enabling comparability. In addition, the replication study maintains a similar methodological structure to the reference study, including the same equipment type, menstrual signal collection, and self-report strategy.

Through the above argumentation, this project thoroughly looks into the boundaries and generalizability of the theoretical effects observed in the reference study, demonstrating the feasibility of using wearable device signals for cycle tracking analysis across different populations..

- Hypothesis: the null hypothesis states that there is no systematic difference in Fitbit-derived physiological or behavioral signals (e.g., resting heart rate, step count, sleep duration) across menstrual cycle phases.

2.1.2. Phase 2: Menstrual Phase Prediction

Following the conceptual replication, the study will expand to capture stable predictive relationships using the full spectrum of data available in the mcPHASES dataset. Machine learning models will be constructed to test whether the combined signals can serve as a reliable basis for personalized applications, such as menstrual-cycle phase detection or family-planning support. We will incorporate a comprehensive feature set including signals from wearable sensors, hormones, and glucose.

2.2. Data Sources

The dataset used in this study is the mcPHASES (Menstrual Cycle Physiological, Hormonal, and Self-reported Events and Symptoms for Menstrual health tracking with Wearables) dataset, version 1.0.0, publicly available through the PhysioNet Platform.[1] This dataset contains a multimodal collection of

health data acquired from 42 participants over observation periods from 2022 to 2024. As mentioned in the study design, only the 2022 interval is used in this study to better align with the original study. Data was acquired from four primary tracks:

- FitBit Sense smartwatch for continuous physiological measurements (e.g. heart rate, body temperature, sleep quality, exercise-related features) (for replication study and ML)
- Dexcom G6 Continuous Glucose Monitor for Glucose data (for ML only)
- Mira Plus Starter Kit (a hormone analyser) for daily biochemical assessments (e.g. luteinizing hormone (LH), Estrone-3-Glucuronide (E3G), progesterone (PdG)) (for ML only)
- Daily diary: self-reported menstrual symptoms and menstruation status (for replication study and ML)

The raw data consists of four independent streams related to the devices used. The physiological data includes high-frequency measurements of heart rate, skin temperature, and activity level gathered by the smartwatch and the glucose feature from the Dexcom G6 monitor. The hormonal data stream provides daily quantitative levels of key reproductive hormones, including LH, E3G, and PdG. The final stream is from self-reported symptoms (e.g., cramping, fatigue) recorded via a mobile application.

2.3. Data Preprocessing

As the study contains two phases, the data preprocessing steps differ between them. The general procedure used is as follows:

We generalized selected sensor-derived tables by aligning records at the participant-day level. The selected datasets (heart rate, exercise, sleep, temperature) were merged using unique participant identifiers (one record per participant per day) and daily timestamps (day in interval) to establish a consistent

temporal structure. This step ensured that each row represented a complete set of physiological indicators for a given day.

The following table shows the general distribution of the data.

Demographics of study participants	**Values**
Age (years), mean (SD)	20.7 (2.6)
Gender, n (%)	
Woman	37 (88)
Non-binary	2 (5)
Gender Fluid	1 (2)
Other	1 (2)
Prefer not to say	1 (2)
Cycle regularity, n (%)	
Irregular	7 (16.7)
Days since last period, median (IQR)	12 (6, 19)

Table 1. Study participant demographics (n = 42).

After getting the demographic information of the study participants, we removed irregular cycles to better align with the reference study.

2.3.1. Data Preprocessing for Replication Study

To replicate the cycle-aligned analytical logic of the reference study, we constructed a daily cycle index for each participant. Using self-reported cycle start dates and the criteria from the original table: the Menstrual phase (MP; days 1 to 5), the Ovulation day (OD; day 1 of the next MP minus 14)[2], the Follicular phase (day 1 of MP to OD minus 1), the Luteal phase (OD plus 1 to the menses), and the Miid-Luteal phase (OD plus 3 to OD plus 9)[2], each day was assigned a cycle day number and mapped to one of the predefined phases (see Appendix A). The original cycle phase labels were contained in the dataset, enabling direct comparison with the theoretical patterns examined in the reference study. In addition, there are slight differences between the daily symptoms for the two studies. For example, in the

reference study, the symptoms are “has_period_started_today”, “bleeding_severity”, “anger_severity”, “anxiety_severity”, “confusion_severity”, “dejection_severity”, “irritability_severity”, “seclusion_severity”, “bloating_severity”, “tenderness_severity”, “headache_severity”, “joint_pain_severity”, “swelling_severity”, and “weight_gain_severity”. However, in our dataset, we have some similar objects, and for other objects, we don’t have any information. So, we chose all the objects that can be a substitute for the objects in the reference paper, which are “flow_volume”, “headaches”, “cramps”, “sore breasts ", “fatigue”, “moodswing”, “indigestion”, and “bloating”.

2.3.1. Data Preprocessing for ML

Building on the participant-day alignment described above, we aggregated the full spectrum of sensor data for machine learning analysis. Given that raw measurements varied in temporal resolution—from per-minute readings (heart rate, glucose) to daily summaries (sleep scores) we applied aggregation methods tailored to each variable's semantic nature. Cumulative metrics such as daily caloric expenditure and step counts were summed across intraday measurements. Continuous physiological signals, including heart rate, glucose levels, and wrist temperature deviation from baseline, were aggregated using daily means, minimums, maximums, and standard deviations to capture both central tendency and variability. Daily cumulative scores, such as stress, retained their final daily value. Exercise data was expanded to all participant-days by imputing zeros for non-exercise days. To prevent data leakage, missing values were handled via forward-fill followed by backward-fill within each participant's time series. Participants with more than 50% missing observations were removed. The resulting dataset contained 2,970 observations across 59 features and 33 participants, with the four-class menstrual cycle phase as prediction target.

To evaluate the impact of feature selection on predictive performance, we employed multiple selection strategies to generate candidate feature subsets. We first established a baseline by removing highly collinear variables (VIF > 10) from the full feature set, addressing redundancy among sleep sub-scores and glucose measures. We then applied four selection methods: ANOVA F-tests to rank features by

between-class variance, SelectKBest to identify the top statistical predictors, and sequential forward and backward selection to optimize cross-validated accuracy through iterative feature addition or removal. The resulting subsets varied in size and composition (up to 52 features and down to 8 features). Each of the five feature configurations was evaluated across all classification algorithms, enabling systematic comparison of how different selection strategies influenced model performance.

2.4. Methods for the Replication Study for Feasibility

We performed a conceptual replication of the FEMFIT study,[2] which examined the feasibility of using wearable Fitbit technology along with digital questionnaires for menstrual cycle tracking. This was a prospective observational cohort study that used retrospective data from the mcPHASES dataset. Participants were instructed to wear their Fitbit device continuously, including at night. Mira Plus Starter Kits and Dexcom G6 CGM were used to analyze hormone levels and glucose levels, respectively. Lastly, participants completed daily questionnaires to report their symptoms throughout the menstrual cycle.

The following table presents a comparison of study characteristics between the Canadian study and the German reference study.

Characteristic	**mcPHASES (Canada)**	**FEMFIT (Germany)**
Enrollment & Retention		
Participants enrolled	50	50
Participants with data analyzed	42	42
Early dropout (within 14 days)	0/42 (0%)	9/42 (21.4%)
Cycle Characteristics		
Average cycle length (days)	27.89 (mean)	28.0 (median)
Cycles included in analysis	$\leq$35 days	$\leq$35 days
Data Completeness		
Partial/incomplete cycles analyzed	116	114
Complete cycles analyzed	84	50
Completed questionnaires (% of days)	89.78%	65.8%

Table 2. Comparison of characteristics: mcPHASES vs. FEMFIT

From the Fitbit devices, we extracted resting heart rate (RHR), daily step counts, and sleep metrics. All cycles were aligned based on the first day of the menstruation phase to standardize comparisons across participants and to mimic the FEMFIT study. Our dataset, which is at the daily level, consisted of 817 observations in the Ovulatory phase, 946 in the Follicular phase, 1,235 in the Luteal phase, and 699 in the Menstrual phase. The FEMFIT study collected 2,468 symptom reports during the Luteal and Menstrual phases, whereas the mcPHASES dataset only had 1,934 symptom reports.

2.4.1. Statistical Analysis

Tools such as Python (version 3.10.4), pandas (1.4.2), matplotlib (3.5.2), seaborn (0.11.2), and numpy (1.22.4) were used for all analyses. Statistical analyses were designed to mimic those performed in the FEMFIT study. Several descriptive statistics were used to assess the feasibility of continuous menstrual cycle data collection. We looked at the mean study participation duration as well as drop out rates within the first 14 days of the study, according to the FEMFIT paper's definition of early dropout. To assess retention, we computed a basic survival analysis curve to show the distribution of days until a participant's last contribution of data. We also looked at data missingness and the percent of study days with valid data from the wearable Fitbit device and daily questionnaires.

Mann-Whitney tests for comparing metrics such as heart rate and sleep variables across menstrual phases. The primary replication target was showing that the average resting heart rate is significantly higher in the Luteal phase compared to the Menstrual phase. Secondary analyses considered step count and sleep duration throughout the cycle. To replicate FEMFIT's findings on the relationship between physical activity and PMS severity, we compared mean daily step counts during the Luteal and Menstrual phases using the Mann-Whitney test.

Lastly, we created 3 mixed-effect models to account for any repeated measures within participants. The regression coefficients and their corresponding p-values were reported and statistical significance was assessed at $\alpha = 0.05$. The following models were fitted for resting heart rate:

- *Model 1: (RHR ~ Phase)* : Fixed effects of Menstrual cycle phases and RHR
- *Model 2: (Steps ~ Phase)*: Fixed effects of Menstrual cycle phases and Steps
- *Model 3: (Minsinsleep ~Phase)*: Fixed effects of Menstrual cycle phases and Sleep duration

2.4.2. Evaluation Metrics

A successful conceptual replication of the FEMFIT study required statistical significance in agreement with the original findings and effect sizes of similar magnitude with acknowledgement to the differences in population, geography, and measurement protocol differences between the German FEMFIT cohort and the mcPHASES dataset.

2.5. Methods for Menstrual Phase Prediction with Machine Learning

We trained four classification algorithms to predict menstrual cycle phase as a four-class outcome (Menstrual, Follicular, Ovulatory, Luteal). Model selection was guided by two considerations: the mixed nature of our dataset, which combines continuous physiological signals, ordinal symptom scores, and categorical demographic variables, and prior work demonstrating the effectiveness of these approaches for menstrual cycle prediction.[8] Each classifier was evaluated on all five feature subsets derived from the feature selection process.

Numeric features were standardized using z-score normalization, while categorical variables were one-hot encoded. Missing values in numeric columns were imputed with zeros, and missing categorical values were assigned a distinct "missing" category. The dataset was split at the observation-level into 80% training and 20% test sets using stratified sampling to preserve class distribution.

2.5.1. Class Imbalance

As detailed in the table below, the dataset exhibits mild class imbalance, with the Menstrual phase accounting for only 18.7% of observations, compared to 32.1% for the Luteal phase. Given the unequal

distribution of cycle phases, all models were configured with balanced class weighting to penalize misclassification of minority classes proportionally to their underrepresentation.

Phase	**Count (n)**	**Percentage (%)**
Luteal	953	32.1
Follicular	805	27.1
Fertility	656	22.1
Menstrual	556	18.7
Total	2,970	100

Table 3. Distribution of target class (menstrual phases)

2.5.2. Classification Algorithms

We used Decision Trees as an interpretable baseline, and Random Forests extended this approach through ensemble learning, aggregating predictions from multiple decision trees trained on bootstrap samples to reduce variance and improve generalization. We also used Logistic Regression with multinomial formulation as a linear baseline and XGBoost to sequentially train weak learners to correct residual errors from prior iterations.[9]

2.5.3. Evaluation Metrics

Model performance was assessed using accuracy, F1 score, and macro-averaged AUC. Macro averaging was chosen to weight all four phases equally, regardless of class frequency, ensuring that performance on minority classes was not obscured by majority class accuracy.

2.5.4. Hyperparameter Tuning

Based on the comparative performance analysis, we identified two distinct model configurations for hyperparameter optimization. First, we selected the top-performing configuration—XGBoost trained on

the Baseline (All 52 Features) set—as the primary candidate for maximizing classification accuracy, serving as the benchmark for the upper limit of predictive capability in this study.

Second, we selected the Random Forest model utilizing Forward Selection features as a lightweight alternative. Despite operating on a significantly reduced feature subset (29 features versus ~52), this configuration retained approximately 95% of the top model's predictive performance. This selection prioritizes practical feasibility, offering an interpretable alternative suitable for resource-constrained wearable applications where minimizing data collection burden is critical.

2.5.5. Adding Temporal Context

To capture the sequential dynamics of the menstrual cycle, we expanded the feature space beyond daily snapshots by incorporating temporal context. We generated lag features (1-day shifts) and 3-day rolling averages for key physiological signals, including nightly temperature, heart rate (mean and minimum), step count, and sleep efficiency. This approach enabled the models to detect phase-dependent trends—such as the gradual rise in basal body temperature during the Luteal phase—rather than relying solely on absolute daily values.

2.6. Ethical Considerations

Our research relies on the ethical approval and data protection protocols established by the original researchers. The original mcPHASES study was approved by the Research Ethics Board at Simon Fraser University and the data was published under the Open Data Commons Attribution License. 42 out of 50 participants in the original research provided informed consent for data collection and the public domain. Data access and use followed the terms of the PhysioNet Data Use Agreement and no new personally identifiable information was collected in the analysis.

3. Results

3.1. Replication Study Results

The conceptual replication aims to verify the existence of systematic differences in Fitbit-derived physiological signals across menstrual cycle phases, applying the analytical framework of the original reference study to the PhysioNet mcPHASES dataset.

3.1.1. Survival Analysis

3.1.1.1. Retention Analysis

As shown in Appendix B, participant retention was evaluated using survival analysis techniques with weekly continuation in the study treated as the outcome. Retention remained at 100% through the first five weeks of data collection, indicating that there is no early dropout. A subtle decline occurred between weeks 6 and 10, after which the retention rate stabilized at 95% for the remainder of the 12-week monitoring period. Overall, the retention rate shows consistent participation over the full study duration.

3.1.1.2. User Engagement Analysis

According to Appendix C.1., our study demonstrates a higher questionnaire completion rate (median = 95%) compared to the reference study. This dramatic improvement may reflect better engagement strategies used in the Canadian cohort or stronger participant commitment to the research objectives. This result is also revealed in the wearable data donation plot (Appendix C.2.). The majority of the participants contribute in the interval of [80, 90] days in the wearable data donation. This indicates that the Canadian cohort has a higher participation rate than the German cohort.

3.1.2. Physiological Signals Analysis

3.1.2.1. Resting Heart Rate Analysis across Phases

The analysis of daily RHR was conducted using a Linear Mixed-effects model (LMEM), accounting for the non-independence of repeated measurements by including participants as a random effect. According to Appendix D.1., the LMEM results confirmed a statistically significant main effect of the menstrual cycle phase on daily RHR (p-value <0.001), leading to the successful rejection of the null hypothesis. The key indicators are: in the Follicular phases, RHR was significantly lower than RHR in the Menstrual phase. The estimated coefficient was 0.736 bpm ($p < 0.001$), indicating a decrease related to the Menstrual phase after controlling for covariates. In the Luteal phase, RHR was significantly higher than RHR in the Menstrual phase. The estimated coefficient was 2.067 bpm ($p < 0.001$), indicating an increase relative to the menstrual phase.

The statistical significance of the phase effect successfully replicates the established biological pattern of heart rate variation observed in the literature, confirming the feasibility of the methodological approach on an independent dataset.

3.1.2.2. Daily Step Count Analysis across Phases

According to Appendix D.2., the analysis of daily average step count indicated that the main effect of the menstrual cycle phase was not statistically significant ($p > 0.1$). Although the coefficient for the Luteal Phase showed a positive correlation(Coef = 411.391), suggesting a slight increase in steps compared to the menstrual phase, this increase failed to reach the 0.05 significance level. So, this project was able to replicate the systematic variation in step count observed in the reference study at the aggregate population level.

3.1.2.3. Sleep Duration Analysis across Phases

According to Appendix D.3., the analysis of sleep duration indicated that the main effect of the menstrual cycle phase was not statistically significant ($p > 0.1$). The coefficient for the Luteal phase showed a positive trend (Coef = 0.248), suggesting a slight increase in sleep duration compared to the Menstrual phase; however, this increase failed to reach the 0.05 significance level. Consequently, this project was able to replicate the systematic variation in sleep duration observed in the reference study at the aggregate population level.

The representation of physiological trends across menstrual cycles are shown in Appendix E.

3.1.2.4. PMS-stratified Daily Step Count Analysis

Following the reference study's analysis, we examined whether physical activity levels differed between participants with varying PMS severity during the Luteal phase and menses. Participants were classified into high PMS (n = X participants; reporting symptom severity ≥ 3 on at least 2 occasions) and no/low PMS (n = Y participants; never reporting severity > 1) groups based on self-reported symptom ratings across all symptom categories during symptomatic phases.

Contrary to the reference study's findings, Mann-Whitney U tests revealed no statistically significant difference in mean daily step counts between PMS severity groups (High PMS: mean = 7,891 steps, SD = 2,292; Low/No PMS: mean = 6,339 steps, SD = 2,641; $U = 169.0$, $p = 0.1223$). Despite the moderate effect size ($r = 0.242$), this difference failed to reach the significance threshold of $\alpha = 0.05$. Notably, the observed trend was directionally opposite to the reference study's finding, where participants with severe PMS walked substantially fewer steps (mean 10,283, SD 6,277) compared to those with no/low PMS (mean 11,694, SD 6,458; $p < 0.001$). In our Canadian cohort, the high PMS group exhibited numerically higher step counts, though this pattern was not statistically reliable and may reflect chance variation,

differences in participant demographics, or regional lifestyle factors that modify the relationship between PMS severity and physical activity. Details are shown in Appendix F.1.

3.1.2.5. PMS-stratified Sleep Duration Analysis

We further examined whether sleep duration differed between participants with varying PMS severity during the Luteal and Menstrual phases, using the same PMS group classifications as the step count analysis. Mann-Whitney U tests revealed no statistically significant difference in mean nightly sleep duration between PMS severity groups (High PMS: mean = 5.69 hours, SD = 0.91; Low/No PMS: mean = 6.23 hours, SD = 0.89; $U = 91.0$, $p = 0.3026$, effect size $r = 0.164$). Although the high PMS group exhibited a trend toward shorter sleep duration (approximately 32 minutes less per night), this difference did not reach statistical significance at the $\alpha = 0.05$ threshold. This finding is consistent with the reference study's observation of no significant relationship between sleep duration and PMS severity ($p = 0.18$), suggesting that while severe PMS symptoms may influence subjective sleep quality, they do not consistently affect objectively measured total sleep time as captured by wearable devices. The small-to-moderate effect size ($r = 0.164$) suggests that any relationship between PMS severity and sleep duration in our cohort was minimal and likely not clinically meaningful. Details are shown in Appendix F.2.

3.1.3. Symptom Analysis

Using Mann-Whitney U tests, we examined whether self-reported symptom severity differed between the Luteal and Menstrual phases. Cramp severity showed a statistically significant difference between phases, with higher severity during the Menstrual phase compared with the Luteal phase ($p = 1.58\times10^{-5}$). This pattern aligns with the reference study. For all remaining symptoms—headache, breast soreness, fatigue, mood swings, indigestion, and bloating, the phase differences were not statistically significant (all $p > 0.05$). Headache ($p = 0.0898$) and breast soreness ($p = 0.0702$) displayed marginal trends toward higher severity during the Menstrual phase, but these did not reach the typical threshold of significance. The

results suggest that cramping is the only symptom exhibiting a reliable phase-dependent increase, while other symptoms show either mild or minor variation between the Luteal and Menstrual phases. These results are very different from the results in the reference study, with more symptoms being insignificant.

Symptom severity score comparison bar plots are contained in Appendix G.

3.2. Machine Learning Prediction Results

3.2.1 Baseline Model Performance and Feature Selection Impact

We evaluated four classification algorithms across five distinct feature subsets to establish a performance baseline before hyperparameter tuning. As detailed in Appendix H, XGBoost consistently outperformed all other classifiers, achieving the highest overall accuracy of 69.36% and a Macro AUC of 0.89 when trained on the full feature set. Random Forest followed closely, reaching 68.18% accuracy on the same set, while Logistic Regression and Decision Trees lagged significantly behind, with peak accuracies of 59.60% and 54.21%, respectively.

Regarding feature selection, while the baseline configuration yielded the highest raw performance, forward selection proved to be the most effective feature selection strategy. It allowed the Random Forest model to maintain a competitive accuracy of 65.99% (Macro AUC: 0.87) despite using a significantly smaller feature set. Conversely, univariate filter methods like SelectKBest resulted in the lowest performance across all models, highlighting the necessity of capturing complex, multivariate interactions in physiological data.

3.2.2 Hyperparameter Tuning Result

For XGBoost, the optimized configuration (n_estimators=300, learning_rate=0.1, max_depth=5, subsample=0.8) achieved a test accuracy of 69.87% and Macro AUC of 0.89, representing a modest 0.5 percentage point improvement over the untuned baseline (69.36%). The Random Forest model, optimized

with n_estimators=300, max_depth=None, and min_samples_split=2, maintained stable performance at 65.66% accuracy (Macro AUC: 0.87), demonstrating high robustness with minimal sensitivity to hyperparameter selection. These marginal gains from tuning suggested that both models had reached a performance plateau when relying solely on daily physiological snapshots, indicating that algorithmic refinement alone would not substantially improve predictive accuracy.

3.2.3 Impact of Temporal Context

To overcome the performance plateau, we incorporated temporal features by adding 1-day lag variables and 3-day rolling averages for six key physiological signals (nightly temperature, heart rate mean and minimum, steps, calories, and sleep efficiency), generating 12 additional features.

The incorporation of temporal context yielded substantial improvements across both model configurations. The XGBoost model trained on Baseline features with temporal enhancements achieved a test accuracy of 70.37% and Macro AUC of 0.90, surpassing both the original baseline (69.36%) and the hyperparameter-tuned baseline (69.87%) with a 1.0 percentage point gain. Notably, the Random Forest model with Forward Selection features showed an even more pronounced improvement, jumping from 65.99% to 68.52% accuracy (Macro AUC: 0.88).

3.2.4 Phase-Specific Classification Performance

Phase	Precision	Recall	F1-Score	Support
Fertility	0.68	0.56	0.62	131
Follicular	0.67	0.73	0.70	161
Luteal	0.72	0.80	0.76	191
Menstrual	0.76	0.67	0.71	111
Macro avg	**0.71**	**0.69**	**0.70**	**594**
Weighted avg	**0.70**	**0.70**	**0.70**	**594**

Table 4. Phase-specific classification performance

Phase-specific performance metrics for the best-performing model (XGBoost with Baseline features and temporal context) revealed notable variation in detection accuracy across menstrual phases. The Luteal phase demonstrated the highest recall (0.80), indicating that the model reliably identified 80% of Luteal phase days. This strong performance likely reflects the distinct physiological signature of the post-ovulatory window.

In contrast, the Ovulatory phase proved most challenging to classify, achieving only 0.56 recall despite reasonable precision (0.68). This difficulty stems from the physiological subtlety of the fertile window – unlike the abrupt onset of menstruation or the thermal shift marking the Luteal phase, the hormonal changes preceding ovulation (rising estrogen and LH surge) produce signals that are often indistinguishable from the late Follicular phase using wrist-based sensors alone. The Menstrual phase achieved moderate recall (0.67) but the highest precision (0.76), while the Follicular phase showed balanced performance (recall: 0.73, precision: 0.67).

4. Discussion

4.1. Interpretation of findings

4.1.1. Interpretation of Replication Study

To demonstrate the same feasibility metrics compared to the reference study, we will validate the scalability of combining wearable questionnaire approaches in the Canadian context. Enhanced retention and data completeness were shown in our study. In addition, the reference study reported considerable challenges with continuous wearable data collection, particularly for sleep metrics, whereas our Canadian participants demonstrated more consistent device adherence. These feasibility improvements confirm that the methodological framework is not only technically solid but also practically implementable across diverse populations with appropriate participant support and engagement protocols. The combination of

high retention and data completeness establishes confidence in the dataset's fitness for subsequent machine learning analyses.

Our conceptual replication research successfully tests the main hypothesis that Fitbit-derived physiological signals exhibit systematic variation across menstrual cycle phases in a Canadian cohort. The statistically significant difference in resting heart rate (RHR) across cycle phases directly replicates the key finding from the German reference study, illustrating that this physiological pattern is strong and stable across geographically and culturally different populations. Specifically, we observed a mean increase of bpm in the Luteal phase relative to menstruation which is consistent with the reference study's reported elevation in the mid and late Luteal phases. This consistency provides strong evidence for the biological validity of using consumer-grade wearable devices to track menstrual cycle-related physiological changes. Additionally, consistent with the reference study's findings, we observed no significant variation in step counts and sleep duration across menstrual cycle phases, successfully replicating their null result. This cross-study consistency suggests that while subjective sleep quality may vary with hormonal changes, measured total sleep time remains relatively stable across the menstrual cycle in healthy populations.

While we successfully replicated core physiological patterns, our study could not replicate the PMS stratified step counts' negative correlation from the reference study, revealing important nuances about cross-regional generalizability. Our PMS-stratified analysis failed to replicate this finding, and notably, the directional trend in our data was opposite to the reference study's observation, with high PMS participants showing numerically higher step counts. In addition, the symptoms analysis is also different. These divergences reflect inherent regional differences in physical activity patterns, coping strategies for menstrual symptoms, or participant demographics rather than methodological failure.

4.1.2. Interpretation of Machine Learning Phase Prediction

Our machine learning analysis revealed that predictive accuracy is driven less by algorithmic complexity and more by explicit temporal modeling. Hyperparameter optimization yielded marginal gains (+0.5%), indicating a performance plateau with static features. The critical breakthrough emerged from incorporating temporal context—1-day lags and 3-day rolling averages—which demonstrated that the slope of physiological signals is a stronger predictor than absolute values. This aligns with established endocrinology: the luteal transition reflects a progesterone-driven temperature shift, not a fixed threshold.

Notably, temporal features provided disproportionately greater benefit to the lightweight Random Forest model (+2.5 percentage points) compared to the complex XGBoost baseline (+1.0 percentage point), demonstrating that resource-constrained architectures can approach state-of-the-art performance when temporal dynamics are properly captured. This enables privacy-preserving, on-device processing without cloud synchronization, addressing key barriers to FemTech adoption.

Phase-specific performance revealed significant variation. The Luteal phase achieved high recall (0.80) due to its distinct thermal and cardiovascular signature, while the Ovulatory phase proved most challenging (recall: 0.56). This difficulty stems from the physiological subtlety of the fertile window; unlike the abrupt onset of menses or the thermal shift marking the luteal phase, pre-ovulatory hormonal changes produce signals often indistinguishable from the late Follicular phase using wrist-based sensors alone, highlighting the need for additional biomarkers such as heart rate variability to enhance fertility detection.

4.2. Limitations & Challenges

The mcPHASES dataset included physiological variables recorded at inconsistent temporal resolutions, ranging from 1–10-minute intervals to daily and mean-level summaries. These discrepancies created challenges for data integration and preprocessing. Following established approaches, we applied several

temporal aggregation strategies to harmonize the data streams and prepare them for machine-learning analysis.

Over 60% of the dataset contained missing or null values, likely due to device malfunction, signal artifacts, or inconsistent device use. To address this, we applied forward- and backward-filling imputation to maintain signal continuity while minimizing distortion of underlying trends.

The small size of our dataset limited the use of deep learning models commonly applied in menstrual-cycle research, which typically require large datasets with over 10,000 cycles and long sequences of more than 10 cycles per participant. Consequently, we focused on more data-efficient methods, including logistic regression, Random Forest, and XGBoost [10,11]

Although our study applies Random Forest and XGBoost, these models do not yield transparent physiological explanations of menstrual phase variability, particularly because irregular cycles were excluded despite being common in real-world populations. While hormone data contribute internally to predictions, their effects are not clearly interpretable at the output level. This limitation underscores the importance of interpretable machine-learning frameworks. Explainability is crucial to examine their interactions, and assess how phase-specific changes in glucose, heart rate, and hormone levels influence cycle classification and alignment with known female hormonal dynamics.

Model-agnostic approaches such as SHapley Additive exPlanation (SHAP) and Local Interpretable Model-agnostic Explanation (LIME) offer tools to address this issue: SHAP attributes feature-level contributions to deviations from average predictions, while LIME produces locally interpretable approximations to explain individual prediction probabilities.[12] SHAP has further demonstrated value in healthcare modelling by improving transparency, interpretability, and clinical relevance.[13] Incorporating these interpretability methods and including participants with irregular cycles represent key next steps to enhance the validity and transparency of future models.

4.3. Implications

4.3.1. Implications for Cross-Regional Signal Validity and Wearable Menstrual Monitoring

The pattern of conceptual replication in our study offers pivotal insights into menstrual cycle biomarker generalizability across populations, revealing a fundamental difference between physiological signals and behavioral or subjective measures with significant implications for wearable-based menstrual health technologies.

Objective physiological parameters from wearable sensors—specifically resting heart rate—demonstrated strong cycle-phase variation replicating consistently across Canadian and German cohorts. This cross-regional stability is theoretically coherent as these signals reflect direct physiological responses to hormonal variations through estrogen and progesterone pathways that are independent of culture, environment, or health behaviors. These routinely grounded biomarkers thus surpass geographic and demographic boundaries.

For developers of consumer wearables like Fitbit (Google), this provides empirical support for implementing menstrual cycle detection algorithms across global users. The cross-regional heart rate consistency implies that machine learning models trained on these physiological features can achieve generalization without large region-specific adjustments, reducing costs and improving scalability.

However, our inability to replicate the reference study's PMS-stratified activity differences, combined with substantially reduced symptom variation in our cohort, reveals critical limitations in behavioral and subjective measure generalizability. Unlike physiological signals directly reflecting hormonal activity, behavioral metrics (step count, symptom reports) are influenced by complex psychosocial, cultural, and environmental factors—including baseline activity patterns, cultural norms, and health literacy—that may influence underlying physiological relationships.

These findings suggest a hierarchical feature architecture for commercial menstrual health applications: physiological signals (heart rate variability) as core algorithmic inputs for cycle phase detection, with behavioral patterns (activity, sleep) and self-reported symptoms as supplementary features enhancing individual personalization. This approach declares that while physiological biomarkers provide a biologically robust foundation for menstrual cycle characterization, behavioral and subjective components require population-specific adjustments or adaptive learning when deployed globally.

4.3.2. Implications for Machine Learning in Cycle Prediction

Our findings suggest that feature design, not model complexity, is the bottleneck in wearable-based menstrual tracking. For device manufacturers, this implies that investments should prioritize time-series preprocessing pipelines over sophisticated cloud-based inference systems.

From a product development perspective, the viability of on-device inference with lightweight models addresses two critical adoption barriers in FemTech: data privacy concerns and battery consumption. Our results provide justification for edge computing architectures where physiological trends are computed locally, with only phase predictions transmitted for cloud storage. This approach aligns with evolving data protection regulations and user preferences for minimizing continuous data transmission.

However, the persistent limitation in detecting the ovulatory window carries significant implications for family planning applications. This shows the need for multi-sensor integration—devices that combine wrist-worn temperature/heart rate with electrodermal activity or intermittent LH testing—to achieve the sensitivity required for conception planning use cases. Until such systems are validated, wearable-only cycle tracking should be positioned for wellness monitoring rather than family planning.

Finally, the finding that temporal context generalizes across different model-feature configurations suggests a robust design principle for future reproductive health algorithms: modeling physiological change over time is universally valuable, independent of specific sensor configurations or algorithmic choices. This principle should guide the development of next-generation menstrual health technologies.

5. Conclusion

This study validated the feasibility of combining wearable sensing and self-reported surveys to characterize menstrual physiology in a Canadian cohort, achieving high data completeness while replicating the core finding of phase-dependent resting heart rate elevation in the Luteal phase. Step count and sleep duration showed no consistent phase effects, and PMS-stratified behavioral and symptom analyses diverged from prior findings, underscoring cohort heterogeneity. Machine-learning results indicated that predictive gains stemmed less from algorithmic complexity than from physiologically informed temporal modeling: baseline XGBoost achieved ~69% accuracy using static features, whereas incorporating lagged variables and rolling averages particularly nightly temperature trends improved accuracy beyond 70% (Macro AUC >0.90). Persistently lower performance for ovulation reflects the subtle physiological signatures of this phase in wearable data. Overall, resting heart rate and temperature dynamics emerge as reliable digital biomarkers, supporting development of interpretable temporal models and inclusion of irregular cycles to enhance transparency, generalizability, and clinical relevance.

6. Acknowledgement

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7. Author Contributions

K.D., R.J., Y.X., and Z.W. contributed equally to study design, data analysis, and manuscript drafting.

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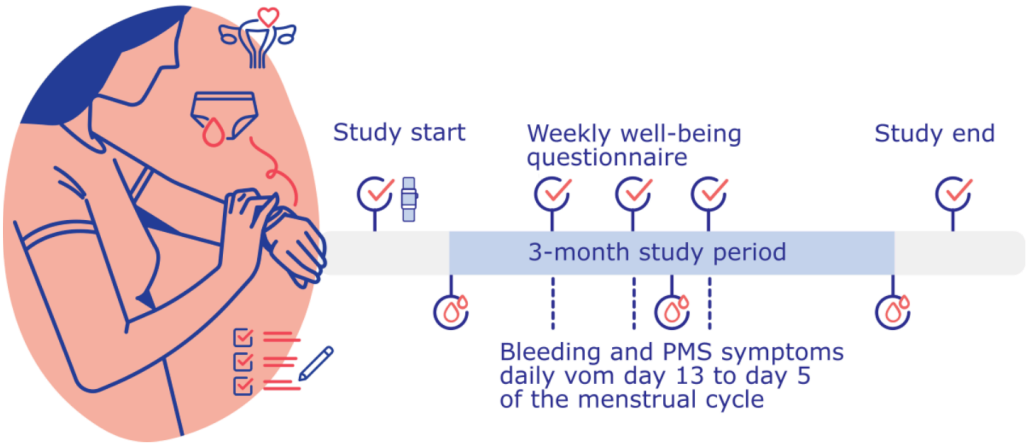
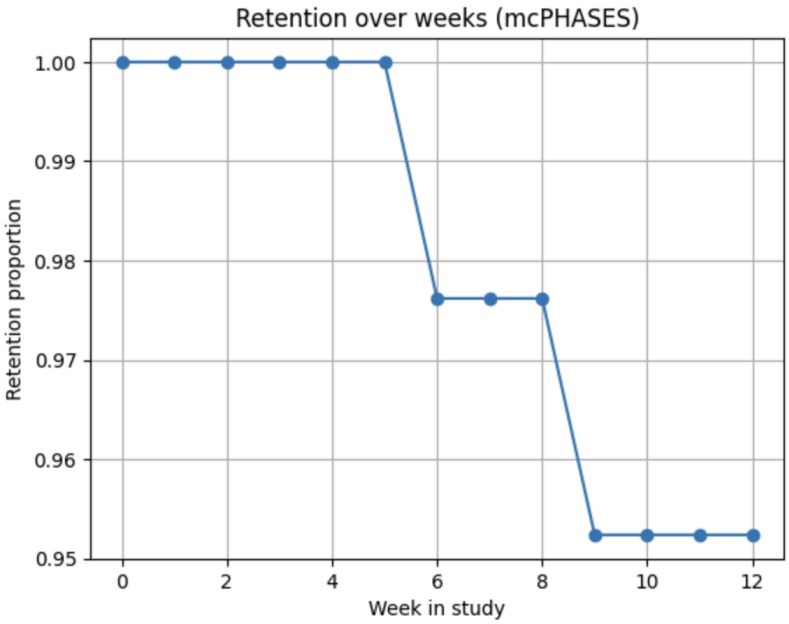
Appendix

Appendix A. Menstrual cycle definition from the Feasibility Study on Menstrual Cycles With Fitbit Device (FEMFIT).

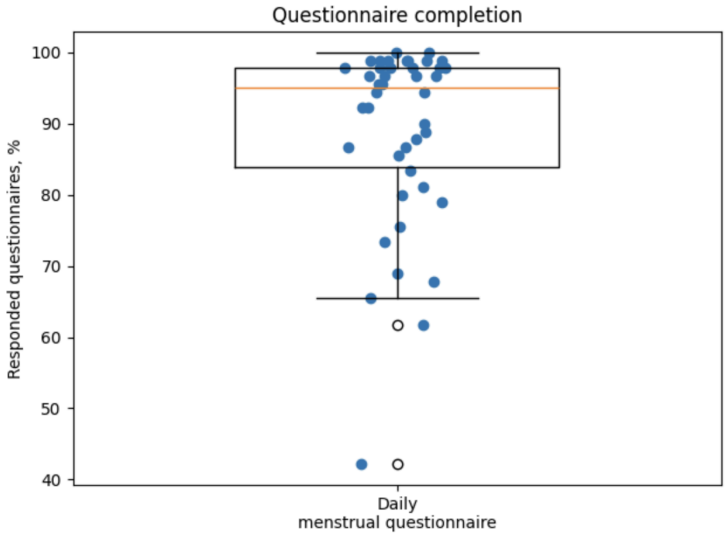


Appendix B. Survival Analysis (Retention Rate Analysis). As shown in the graph, the retention rate for the Canadian cohort never dropped below 95%, indicating a higher retention rate than the German cohort.

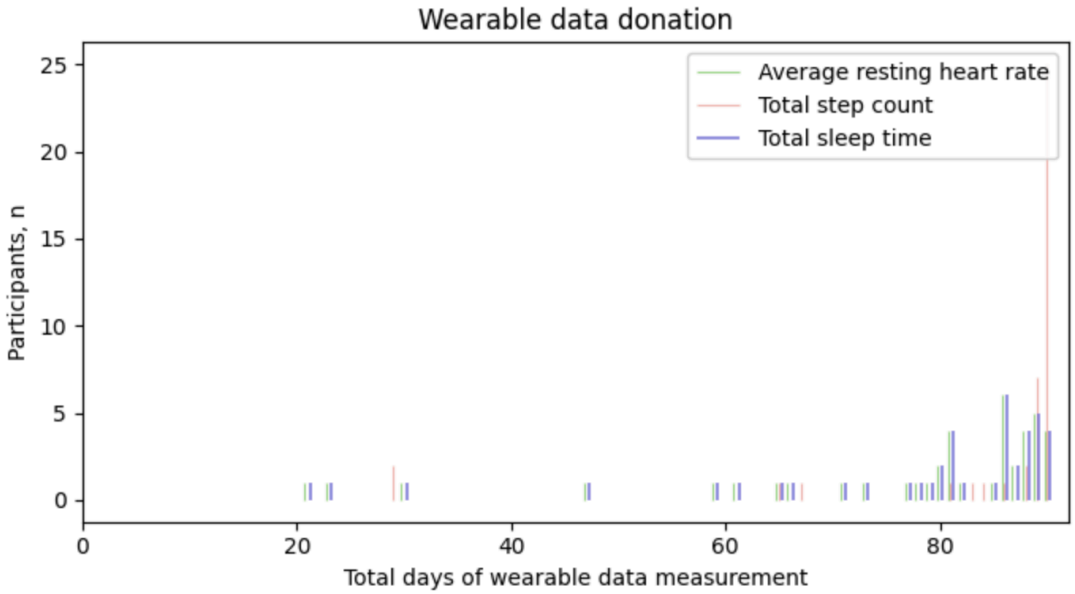


Appendix C. User Engagement Analysis.

C.1. Daily menstrual questionnaire completion box plot. The daily questionnaire is collected once per day. The box plot shows the median and IQR, with whiskers showing minimum and maximum values for outliers. The dots in the graph present the data points.



C.2. Wearable data donation bar plot. Bars' colors are based on different variables.



Appendix D. Mixed Linear Model Regression Results.

D.1. resting_heart_rate

Mixed-effects model estimates for *resting_heart_rate*.

Variable	**Coef.**	**z**	**p-value**	**2.5%**	**97.5%**
Luteal phase	2.067	14.753	< 0.001	1.793	2.342
Menstrual phase	0.736	4.843	< 0.001	0.438	1.033

D.2. Steps

Mixed-effects model estimates for *steps*.

Variable	Coef.	z	p-value	2.5%	97.5%
Luteal phase	411.391	1.438	0.150	-149.305	972.087
Menstrual phase	358.327	1.132	0.258	-261.991	978.644

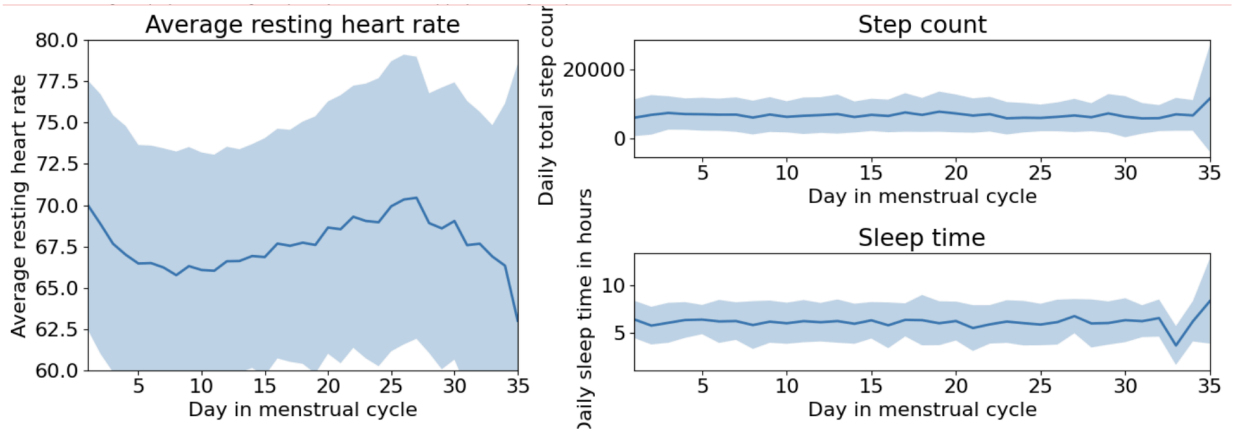
D.3. Sleep duration

Mixed-effects model estimates for *minutesasleep*.

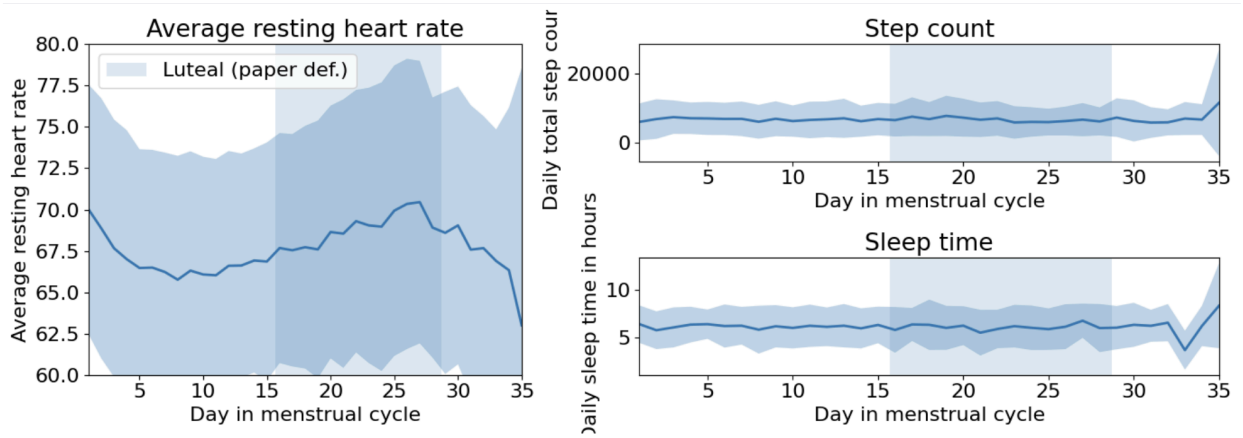
Variable	Coef.	z	p-value	2.5%	97.5%
Luteal phase	3.394	0.359	0.720	-15.141	21.930
Menstrual phase	10.574	1.013	0.311	-9.879	31.027

Appendix E. PMS Severity Relation Analysis. Physiological trend plots. The intervals highlighted are the Luteal phases in the cycles.

E.1. Physiological trend plots with the phases label contained in mcPHASES.



E.2. Physiological trend plots with the phases calculated using the definition in the reference study.



These two graphs show exact same trends and physiological variations between phases, leading to the conclusion that the FITBIT detection of phases is accurate in mcPHASES.

Appendix F. PMS Severity Relation Analysis.

F.1. PMS stratified step analysis

Comparison of Steps Between Severe PMS and Low PMS Groups

Group	Mean ± SD (steps)	U	p-value
Severe PMS	7891 ± 2292	169.0	0.1223
Low PMS	6339 ± 2641		
		Effect size $r = 0.242$	

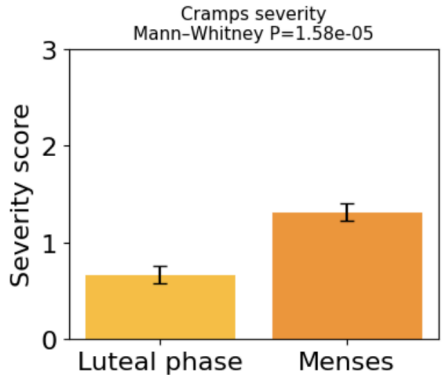
F.2. PMS stratified sleep analysis

Comparison of Sleep Duration Between Severe PMS and Low PMS Groups

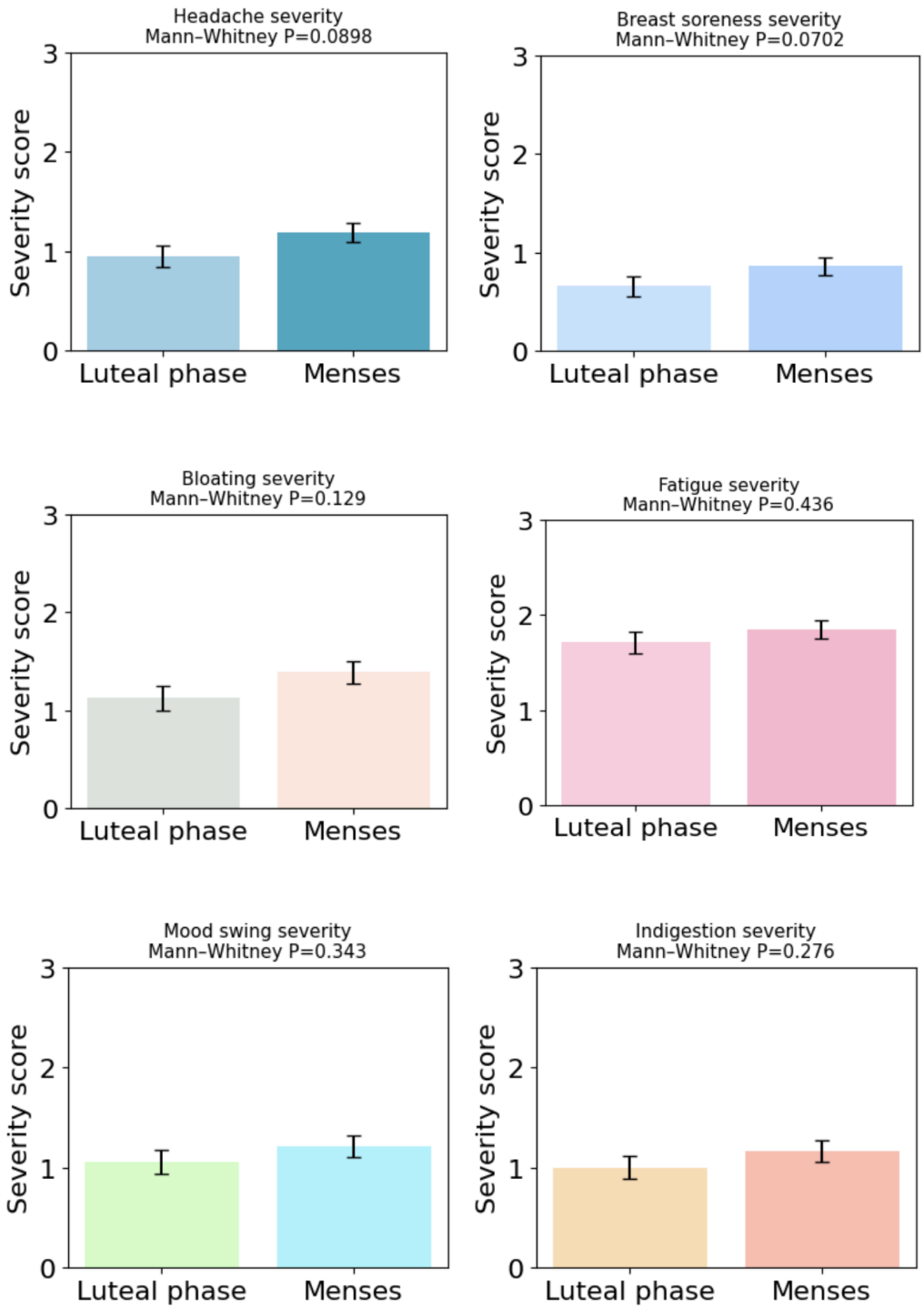
Group	Mean ± SD (hours)	U	p-value
Severe PMS	5.69 ± 0.91	91.0	0.3026
Low PMS	6.23 ± 0.89		
		Effect size $r = 0.164$	

Appendix G. Symptoms Severity Analysis

G.1. Symptoms with significant difference.



G.2. Symptoms with non-significant difference.



G.3. Mann-Whitney tests (Luteal vs Menses) results.

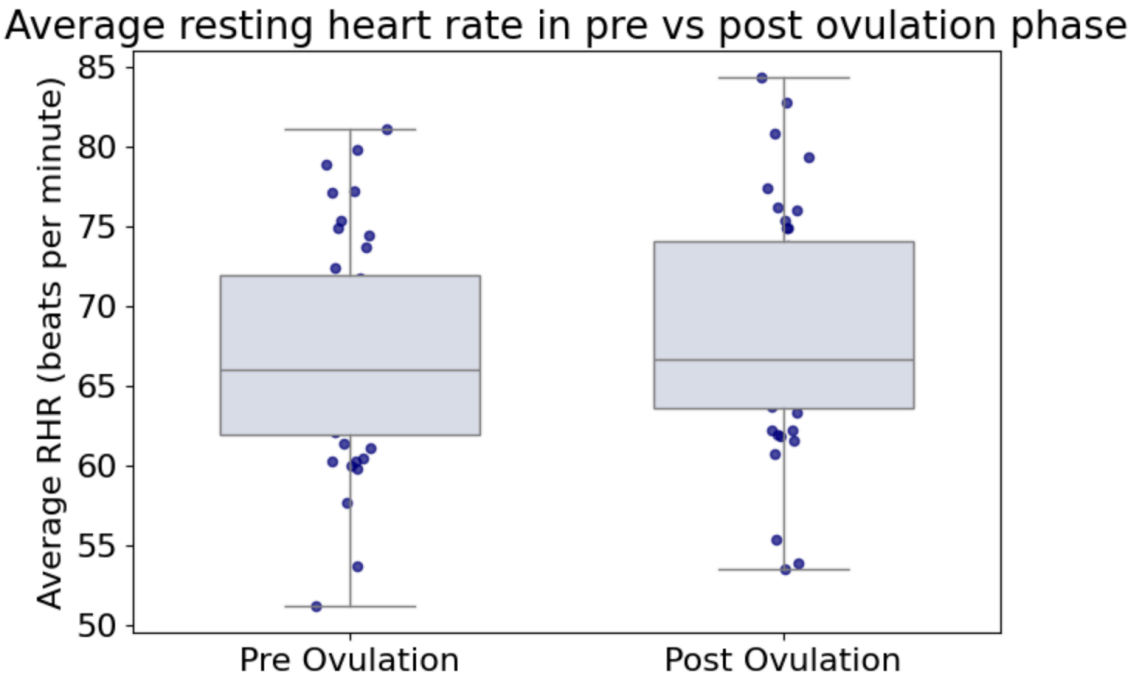
Mann–Whitney tests comparing Luteal vs. Menses phases

Outcome	U	p-value	Mean (L)	Mean (M)	n
Headache severity	692.0	0.08976	0.949	1.191	42
Cramps severity	399.5	1.579×10^{-5}	0.664	1.311	42
Breast soreness severity	680.0	0.07021	0.658	0.862	42
Fatigue severity	794.5	0.4362	1.710	1.851	42
Mood swing severity	775.5	0.3425	1.062	1.212	42
Indigestion severity	760.0	0.2765	1.002	1.167	42
Bloating severity	712.0	0.1291	1.124	1.387	42

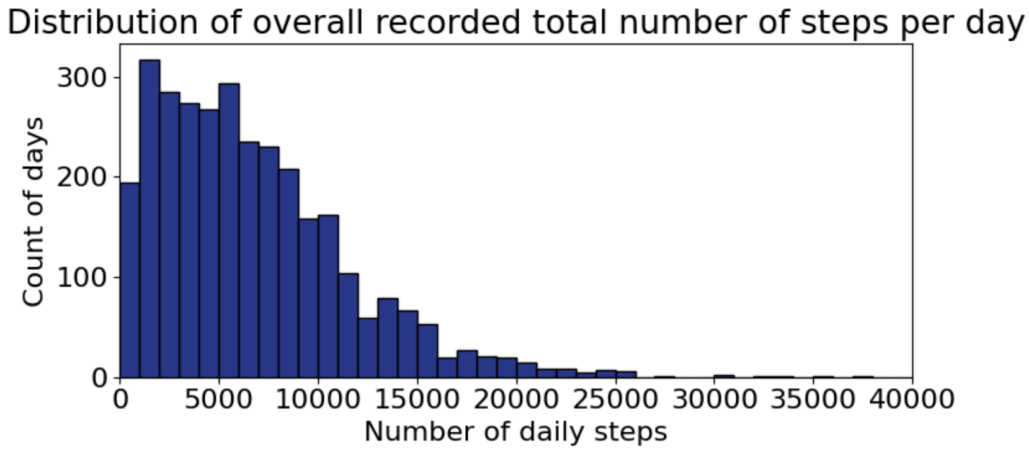
Appendix H. Pre-Tuning Machine Learning Performance

Rank	Feature Set	Model	Accuracy	Macro F1	Macro AUC
1	Baseline (All Features)	XGBoost	0.6936	0.6896	0.8906
2	Baseline (All Features)	Random Forest	0.6818	0.6695	0.883
3	Forward Selection	Random Forest	0.6599	0.6448	0.8697
4	Forward Selection	XGBoost	0.6532	0.6482	0.8732
5	Backward Selection	Random Forest	0.6481	0.6359	0.8574
6	ANOVA	Random Forest	0.6465	0.6394	0.8563
7	Backward Selection	XGBoost	0.6263	0.6231	0.8471
8	ANOVA	XGBoost	0.6195	0.6195	0.8406
9	SelectKBest	XGBoost	0.6044	0.6062	0.8291
10	Baseline (All Features)	Logistic Regression	0.596	0.5965	0.8327
11	SelectKBest	Random Forest	0.5875	0.5807	0.8239
12	SelectKBest	Logistic Regression	0.5724	0.5709	0.8088
13	Forward Selection	Logistic Regression	0.5707	0.573	0.8164
14	ANOVA	Logistic Regression	0.569	0.5655	0.8118
15	Backward Selection	Logistic Regression	0.5589	0.5564	0.8082
16	Baseline (All Features)	Decision Tree	0.5421	0.5423	0.6916
17	ANOVA	Decision Tree	0.5236	0.5248	0.6832
18	Backward Selection	Decision Tree	0.5219	0.5208	0.6813
19	SelectKBest	Decision Tree	0.5118	0.5129	0.6728
20	Forward Selection	Decision Tree	0.4815	0.4833	0.6506

Appendix I. Average RHR comparison between phases. The result shows the same pattern as detected from the reference study.



Appendix J. Distribution of the overall recorded total number of steps per day. The results show the same positive skewed distribution as the reference study.



PMS stratified Step count shows a reverse correlation to the reference study